IGCSE 0607 Practice Questions — E5.1 — Combined Questions & Markscheme

Geometrical terms

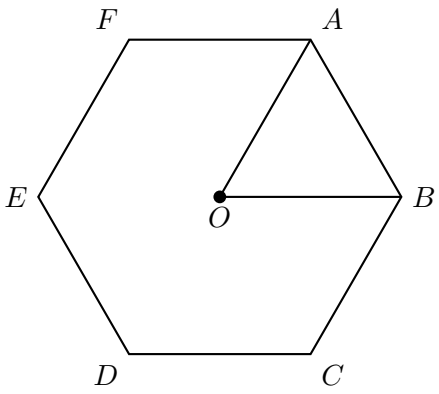
Total Questions: 9 — **Total Marks:** 53 Marks

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  → **Paper 2 (Non-Calculator)** *(6 Qs)* · **Paper 4 (Calculator)** *(3 Qs)*

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1 The diagram shows a regular polygon $ABCDEF$ with centre O.



(a) Write down the mathematical name of the polygon $ABCDEF$.

..... **[1]**

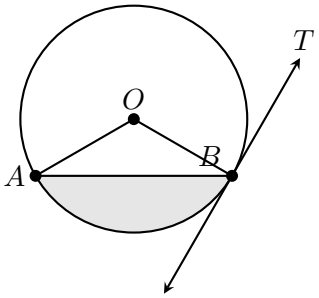
(b) Write down the mathematical name for the angle AFE.

..... **[1]**

(c) Write down the mathematical name for the triangle AOB.

..... **[1]**

2 The diagram shows a circle, centre O.
A and B are points on the circumference of the circle.
The line passing through T touches the circle at B.



(a) Write down the geometrical name of the line AB.

..... **[1]**

(b) Write down the geometrical name of the line BT.

..... **[1]**

(c) Write down the geometrical name of the shaded region.

..... **[1]**

(d) Write down the mathematical name of triangle OAB.

..... **[1]**

3 **(a)** **(i)** A polygon has interior angles that sum to 1080°.

Work out the number of sides of this polygon.

..... **[2]**

(ii) Write down the mathematical name of this polygon.

..... **[1]**

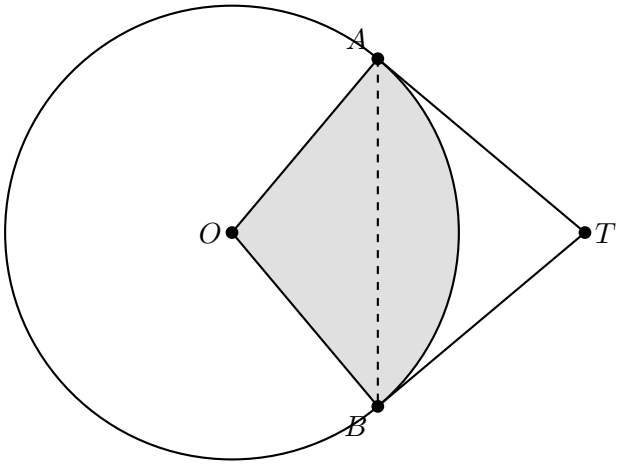
(b) Write down the mathematical name of a quadrilateral with all four sides equal in length and no right angles.

..... **[1]**

(c) Write down the mathematical name of a 3D solid that has a uniform cross-section and two parallel circular faces.

..... **[1]**

4 Points A and B lie on a circle, centre O.
The lines TA and TB are tangents to the circle.
The region enclosed by the arc AB and the radii OA and OB is shaded.



(a) Write down the mathematical name for the quadrilateral $OATB$.

..... **[1]**

(b) Write down the mathematical name for the shaded region.

..... **[1]**

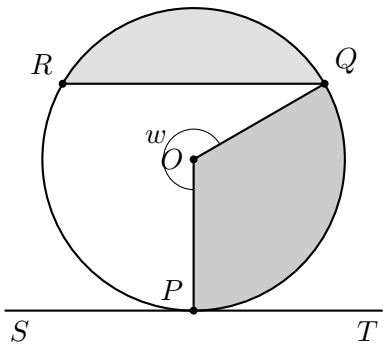
(c) Write down the mathematical name for the straight line AB.

..... **[1]**

(d) Triangle OAT and triangle OBT are exactly the same size and shape.
Write down the mathematical term for two identical shapes.

..... **[1]**

5 The diagram shows a circle, centre O.
P, Q and R are points on the circumference.
The straight line ST touches the circle at P.



(a) Write down the mathematical name of the line ST.

..... **[1]**

(b) Write down the mathematical name of the line segment QR.

..... **[1]**

(c) Write down the mathematical name of the shaded region bounded by the line QR and the minor arc QR.

..... **[1]**

(d) Write down the mathematical name of the shaded region bounded by the radii OP and OQ and the minor arc PQ.

..... **[1]**

(e) The angle marked w is greater than $180°$ and less than $360°$.
Write down the geometrical term for this type of angle.

..... **[1]**

6 **(a)** The interior angle of a regular polygon is 144°.

(i) Work out the number of sides of this polygon.

..... **[2]**

(ii) Write down the geometrical name of this polygon.

..... **[1]**

(b) A hexagon has interior angles of $x°$, $2x°$, $(x + 30)°$, $(3x + 60)°$, 110° and 100°.

(i) Calculate the value of x.

$x =$ ..... **[3]**

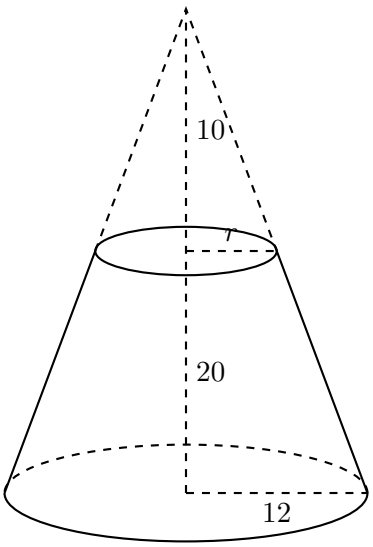
(ii) Write down the geometrical name for an angle of $(x + 30)°$.

..... **[1]**

(iii) Write down the geometrical name for an angle of $(3x + 60)°$.

..... **[1]**

7 The diagram shows a large solid cone with a smaller solid cone removed from the top.
The cut is made parallel to the base of the large cone.
All dimensions are in centimetres.



(a) Write down the mathematical name of the solid with height 20.

..... **[1]**

(b) The small cone is geometrically related to the original large cone.

(i) Write down the mathematical word that describes the relationship between the small cone and the large cone.

..... **[1]**

(ii) Work out the value of the radius, r.

$r =$ ..... **[2]**

(iii) Calculate the slant edge of the solid with height 20.

..... *cm* **[3]**

8 A map shows three towns A, B and C.
Town B is 45 km from A on a bearing of 062°.
Town C is 45 km from A on a bearing of 138°.

(a) Write down the mathematical name of triangle ABC.

..... **[1]**

(b) Work out the bearing of A from C.

..... ° **[2]**

(c) **(i)** Point M is the midpoint of the straight line BC.
Write down the mathematical name of the line AM in relation to the line BC.

..... **[1]**

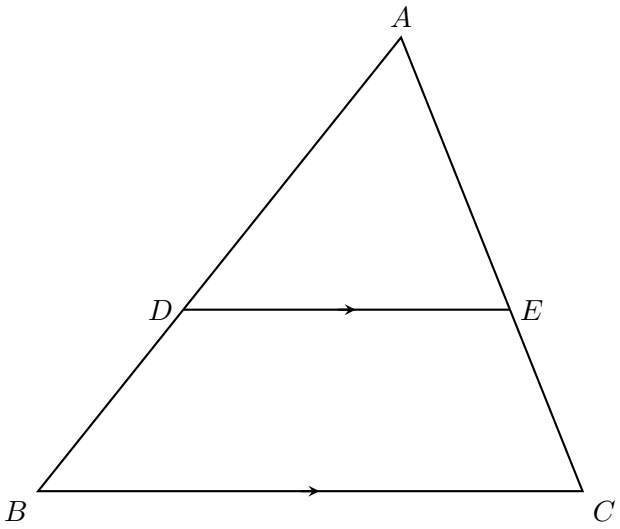
(ii) Calculate the bearing of M from A.

..... ° **[2]**

(d) A fourth town D is located such that the line AB is parallel to the line DC and the line AD is not parallel to the line BC.
Write down the mathematical name of the quadrilateral $ABCD$.

..... **[1]**

9 The diagram shows triangle ABC.
D is a point on AB and E is a point on AC such that DE is parallel to BC.
$AD = 6$ cm, $DB = 4$ cm, $AE = 7.5$ cm and $BC = 15$ cm.



(a) Explain why triangle ADE is similar to triangle ABC.

[2]

(b) Calculate the length of DE.

..... cm **[2]**

(c) Calculate the length of EC.

..... cm **[2]**

(d) State the geometrical name of the quadrilateral $DBCE$.

..... **[1]**

(e) The area of triangle ADE is 18 cm^2.
Calculate the area of the quadrilateral $DBCE$.

..... cm^2 **[3]**

1 **(a)** Hexagon **[B1]**

(b) Obtuse **[B1]**

(c) Equilateral **[B1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Minor segment (Accept segment) **[B1]**

(d) Isosceles **[B1]**

[Back to Question] *[Total Marks: 4]*

3 **(a)** **(i)** $(n-2) \times 180 = 1080$ or $1080 \div 180$ **[M1]**

8 **[A1]**

(ii) Octagon **[B1]**

(b) Rhombus **[B1]**

(c) Cylinder **[B1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** Kite **[B1]**

(b) Sector (or minor sector) **[B1]**

(c) Chord **[B1]**

(d) Congruent **[B1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** Tangent **[B1]**

(b) Chord **[B1]**

(c) Segment **[B1]**

(d) Sector **[B1]**

(e) Reflex **[B1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** **(i)** Exterior angle $= 180^\circ - 144^\circ = 36^\circ$ **[M1]**

$\frac{360}{36} = 10$ **[A1]**

(ii) Decagon **[B1]**

(b) **(i)** Sum of interior angles $= (6-2) \times 180° = 720°$ **[B1]**

$x + 2x + (x + 30) + (3x + 60) + 110 + 100 = 720$ OR $7x + 300 = 720$ **[M1]**

$x = 60$ **[A1]**

(ii) Right angle (since $60 + 30 = 90$) **[B1]**

(iii) Reflex angle (since $3(60) + 60 = 240$) **[B1]**

[Back to Question] *[Total Marks: 8]*

7 **(a)** Frustum **[B1]**

(b) **(i)** Similar **[B1]**

(ii) Scale factor $= \dfrac{20+10}{10} = 3$ or $\dfrac{10}{30} = \dfrac{1}{3}$ **[M1]**

$r = \dfrac{12}{3} = 4$ **[A1]**

(iii) Radius difference $= 12 - 4 = 8$ **[M1]**

$\sqrt{20^2 + 8^2}$ **[M1]**

21.5 or 21.54... **[A1]**

[Back to Question] *[Total Marks: 7]*

8 **(a)** Isosceles **[B1]**

(b) $138 + 180$ or $360 - (180 - 138)$ **[M1]**

318 **[A1]**

(c) **(i)** Perpendicular bisector **[B1]**

(ii) Angle $BAC = 138 - 62 = 76°$ **[M1]**

Bearing $= 062 + (76 \div 2) = 100°$ **[A1]**

(d) Trapezium **[B1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** Angle ADE = Angle ABC because they are corresponding angles **[B1]**

Angle A is common to both triangles OR Angle AED = Angle ACB (corresponding angles) **[B1]**

(b) Length scale factor $= \dfrac{6}{10}$ or 0.6 **[M1]**

$DE = 15 \times 0.6 = 9$ **[A1]**

(c) $AC = \dfrac{7.5}{0.6} = 12.5$ **[M1]**

$EC = 12.5 - 7.5 = 5$ **[A1]**

(d) Trapezium **[B1]**

(e) Area scale factor $= 0.6^2 = 0.36$ **[M1]**

Area of triangle $ABC = \dfrac{18}{0.36} = 50$ **[M1]**

Area of $DBCE = 50 - 18 = 32$ **[A1]**

[Back to Question] *[Total Marks: 10]*